23 In vitro fertilisation

Not all couples that wish to have a baby are able to do so. About 15% of couples will experience difficulty in trying to achieve a pregnancy. There are many reasons why a couple can be considered 'infertile', and sometimes no cause for the infertility can be found.

1 Give two reasons why a male might be infertile.

2 Give two reasons why a female might be infertile.

The infertile couple has options. One of these is *in vitro fertilisation*, or *IVF*, a medical procedure to help fertilisation. *In vitro fertilisation* – literally, *'in-glass'* fertilisation – involves fertilisation of ova outside of the body. Techniques vary, but the following shows a common sequence:

- Fertility drugs are given to the female to stimulate ova production.
- Ova are removed from the ovary by laparoscopic surgery, a minimally invasive technique that only involves making a small incision.
- The ova are placed in a glass container, sperm is added and they are incubated for about 24 hours for fertilisation to take place.
- The fertilised eggs (zygotes) are transferred to a growth medium to ensure they can divide and grow into embryos.
- Embryos that develop normally are identified. A small number (two to four) are placed in the woman's fallopian tube or can be implanted into the womb.
- Embryos not used immediately may be frozen (in liquid nitrogen) for later use.

3 Give two non-medical options for an infertile couple.

4 IVF is a time-consuming and expensive procedure that is not always successful, even after multiple attempts. Discuss why scarce medical resources are used on these procedures when some people have to wait a long time for treatment for life-threatening conditions.

5 Do you think a child should be told that they were conceived using IVF? Explain your answer.

6 Many embryos are created during the IVF process. Frequently there are some 'left over' when a procedure for a couple is successful. What should happen to these embryos?



Louise Brown was the first person to be born using IVF. Two British doctors, Patrick Steptoe and Robert Edwards, pioneered the process used. Louise also had a sister who was conceived using IVF. Louise has been the centre of a great deal of publicity since her birth in 1978. In 2003 she attended a reunion party at the clinic where IVF was first developed. Today the procedure is carried out in many clinics around the world. It can cost around $20 000 to $30 000 for a single IVF cycle with no certainty of conception.

7 Imagine you are Louise Brown. Someone says to you, "Hey, you're the girl who was born in a test tube!". Outline your response to this statement.

8 Discuss whether or not the cost of IVF treatment should be subsidised by government.

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Surrogacy

Surrogate mothers are women who carry the implanted, fertilised zygote of another couple, usually because the biological mother cannot undergo the pregnancy. The surrogate mother commonly has her medical bills paid and receives a final payment after the child is born and given to its parents. Cases have occurred where the surrogate mother changes her mind and wishes to keep the baby rather than accept payment for carrying it (this includes refunding all medical expenses).

9 Do you agree or disagree with the concept of surrogacy? Give your reasons.

10 Give reasons why the surrogate mother might change her mind.

11 Which woman do you consider the 'real' mother?

12 If you were a judge, whose case would you support, the surrogate mother or the biological mother? Why?

Pre-implantation Genetic Diagnosis (PGD)

The term 'genetic screening' is becoming familiar to most people these days. PGD is a technique that works with standard IVF methods to check whether the embryos that are developing for implantation are carrying genes for genetic disorders.

13 Outline what is meant by 'IVF'.

Prior to implantation in the mother, embryos are tested, usually on day three of fertilisation. When a piece of tissue is removed from a living organism, to be examined under the microscope, the procedure is called a *biopsy*. To obtain the material needed to test the embryo, an *embryo biopsy* is performed. This is done by carefully removing a single cell from the embryo and it is believed that the procedure causes no ill-effect to its development. The embryo is returned to the petri dish to continue growing while tests are performed on the DNA within the extracted cell. This will take from several hours to a whole day but results are usually obtained within six hours. If no evidence is present for genetic disorders, the embryo will be one of those implanted in the mother.

14 If an embryo tests 'clean' does this mean a pregnancy will result? Explain your answer.

15 If the embryo tests positive for a disorder, the family is presented with the decision abort it. Pro-life supporters believe that this is no less fair than any other abortion while others feel it is the kinder option from both the parents' and child's point of view. Which view do you support and why?

16 Outline the difference between an 'embryo' and a 'foetus'.

X-linked diseases

A number of X-linked diseases exist and many are associated with muscle disorders or *myopathy* (*myo* muscle, *pathos* disease). While not all of the muscular dystrophies are X-linked, many have at least one X-linked variation. Examples include:

- Duchenne Muscular Dystrophy (DMD)
- Emery-Dreifuss Muscular Dystrophy (EDMD)
- Spinal Bulbar Muscular Atrophy (SBMA)
- Charcot-Marie-Tooth Disease (CMT)
- Myotubular Myopathy (MTM)

17 Some of the conditions above are fatal while others, such as EDMD, are benign. Give a definition for *benign*. Compare it with *malignant*.

18 Describe the difference between a *sex-linked* disease and a *sexually transmitted* disease.

19 Write a paragraph describing one of the X-linked muscular disorders named above.

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